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Abundance from Abroad: Migrant Income and Long-Run Economic Development

Contents

1 Big picture

- **Question.** How do persistent increases in international migrant income prospects affect long-run economic development in migrant-origin areas?
- **Setting.** The paper studies Philippine provinces after the 1997 Asian Financial Crisis. Provinces differed in their pre-crisis distribution of migrant earnings across destination countries, so exchange-rate movements generated heterogeneous shocks to migrant income prospects across provinces.
- **Core challenge.** Migrant income is endogenous: richer provinces may both send more migrants and earn more at home. The paper therefore needs a source of exogenous variation in migrant income that is persistent, geographically heterogeneous, and clearly separable from other channels such as trade.
- **Identification idea.** The authors build a province-level shift-share shock from destination-specific exchange-rate changes (the "shifts") and pre-1997 province-destination migrant income exposure weights (the "shares").
- **Main empirical results.**
  - A one-standard-deviation migrant income shock raises long-run migrant income by about 13.9%.
  - Long-run domestic income rises by about 6.5%.
  - Of the total increase in global income, 74.9% comes from domestic income and 25.1% from migrant income.
  - Household expenditure per capita also rises, and the gains accumulate gradually over roughly two decades.
- **Main mechanisms.** The shock increases education, raises migration rates, shifts migrants into more skilled occupations, raises migrant salaries, and moves local labor away from primary sectors and toward nontradable services. It also plausibly relaxes liquidity constraints and finances domestic enterprise investment.
- **Model-based quantification.** Educational investments account for about 19.7% of the long-run increase in global income. The combination of education, persistent exchange-rate gains, and migration reallocation explains about 89% of the increase in migrant income.
- **Why this paper matters.** The paper moves beyond household-level remittance effects and shows that persistent access to high-income migration opportunities can generate broad origin-area development, with most long-run gains arising from domestic rather than migrant income.

2 Data and measurement (key objects)

2.1 Observed migrant-income exposure data

- **Unit of variation.** Philippine province o, destination country d, and year t.
- **Core innovation.** The paper constructs pre-shock province-destination migrant income per capita, which serves as the exposure weight in the shift-share design.

- **Administrative sources.** Exposure weights are built by matching two Philippine government administrative datasets:
  - OWWA data, which contain migrants' Philippine home addresses,
  - POEA data, which contain migrant income and occupation.
- **Why these data matter.** Standard census or survey datasets do not report the full matrix of migrant income flows from overseas destinations back to subnational origins. Without these administrative data, the paper's shift-share design would not be feasible.
- **Geographic coverage.** The analysis uses 74 Philippine provinces, harmonized to a consistent 1990 boundary definition.

2.2 Observed exchange-rate shocks

- **Source.** Bloomberg exchange-rate data.
- **Shock window.** The destination-specific exchange-rate shock is the proportional change in the average destination-currency-per-Philippine-peso exchange rate from the pre-shock period (July 1996–June 1997) to the post-shock period (October 1997–September 1998).
- **Economic meaning.** A positive shock means an appreciation of the destination currency against the Philippine peso, which raises the peso value of migrant earnings in that destination.
- **Persistence.** A crucial feature of the design is that these exchange-rate movements proved highly persistent over the next two decades.

2.3 Observed outcome data

- **Family Income and Expenditure Survey (FIES).** Used for domestic income per capita and expenditure per capita. Triennial waves from 1985 to 2018 provide up to four pre-shock observations, the partially treated 1997 wave, and up to seven post-shock observations.
- **Migrant income.** Constructed from POEA/OWWA contract data, available for 1994, 1997, 2009, 2012, and 2015 in the main long-run income analysis.
- **Global income.** Defined as domestic income per capita plus migrant income per capita.
- **Census data.** Used for educational attainment, migrant share in population, migrant skill composition, and broad employment shares across sectors (1990, 1995, 2000, 2007, 2010, 2015; sectoral employment shares available for 1990, 1995, 2000, and 2010).
- **Additional outcomes.** The paper also studies average migrant salaries, migrant contracts by occupation, manufactured exports, consumer prices, internal migration, and foreign direct investment.

2.4 Unobservables and timing

- **Province fixed heterogeneity.** Time-invariant province characteristics are absorbed by province fixed effects.

- **Timing.** The exchange-rate shock hits in 1997. The main reduced-form design compares pre-1997 and post-1997 outcomes across provinces with different baseline migrant-income exposure.
- **Persistence.** The long-run treatment is persistent because both its components persist:
  - destination exchange-rate changes remain far from their pre-crisis levels,
  - province-destination migrant exposure weights remain persistent because migration networks and dyad-specific migration costs generate only partial adjustment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Weighted-average exchange-rate shock

The first building block is the average exchange-rate shock affecting province o:

$$Rshock_o = \frac{\sum_d \omega_{do0} \widetilde{\Delta R_d}}{\sum_d \omega_{do0}}. \tag{1}$$

- $\widetilde{\Delta R_d}$ is the destination-d exchange-rate change around the 1997 crisis.
- ω_{do0} is pre-shock migrant income per capita in province o coming from destination d.
- **Interpretation.** $Rshock_o$ is a weighted-average exchange-rate shock, where the weights reflect the province's pre-crisis destination mix of migrant income.

3.2 Pre-shock migrant income per capita

Province-level migrant income per capita before the crisis is

$$MigInc_{o0} = \sum_d \omega_{do0}. \tag{2}$$

- **Interpretation.** Provinces with the same weighted-average exchange-rate shock can still differ in exposure if one province had a much larger baseline migrant-income sector.

3.3 The migrant-income shift-share shock

The key treatment variable is the interaction of the weighted-average exchange-rate shock and baseline migrant income per capita:

$$Shiftshare_o = Rshock_o \times MigInc_{o0} = \sum_d \omega_{do0} \widetilde{\Delta R_d}. \tag{3}$$

- **Shift-share interpretation.** The "shifts" are the destination-specific exchange-rate shocks $\widetilde{\Delta R_d}$, and the "shares" are the province-destination exposure weights ω_{do0}.
- **Economic meaning.** This is the predicted change in migrant income per provincial resident due solely to the destination-level exchange-rate changes, given the province's pre-crisis migrant destination structure.

3.4 Baseline two-way fixed-effects regression

The main estimating equation is

$$\begin{aligned} y_{ot} = {} & \alpha_o + \gamma_t + \beta_1 \, Shiftshare_o \times Post_t \\ & + \lambda'(MigInc_{o0} \times D_t) + \phi'(Rshock_o \times D_t) + \delta'(X_{o0} \times Post_t) + \varepsilon_{ot}. \end{aligned} \tag{4}$$

- y_{ot} is a province-level outcome, such as domestic income per capita, migrant income per capita, expenditure per capita, educational attainment, or employment shares.
- α_o are province fixed effects.
- γ_t are time fixed effects.
- $Post_t$ is an indicator for post-1997 periods.
- D_t is a full set of period indicators used to control flexibly for the main effects of the interaction components.
- X_{o0} is a vector of baseline province controls, including development and industrial-structure variables.
- **Interpretation of β_1.** The coefficient measures how much outcomes change after 1997 in provinces that had larger pre-crisis exposure to destinations with stronger exchange-rate appreciations.

3.5 Event-study specification

To inspect dynamics and pre-trends, the paper replaces the single post-shock interaction with year-specific interactions:

$$y_{ot} = \alpha_o + \gamma_t + \sum_{\tau \neq 1994} \beta_\tau \left(Shiftshare_o \times \mathbf{1}\{t = \tau\} \right) + \text{controls} + \varepsilon_{ot}. \tag{4'}$$

- The 1994 interaction is omitted as the reference year.
- **Interpretation.** This specification checks whether high-exposure provinces were already on different trajectories before the crisis and shows how effects accumulate over time after 1997.
- **Finding.** There are no troubling positive pre-trends, while post-1997 coefficients rise gradually, consistent with persistent and accumulating development effects.

3.6 Import and export shift-share controls

To rule out trade-based confounding, the paper builds analogous shift-share variables for imports and exports:

$$Shiftshare_o^m = \frac{1}{Pop_o} \sum_d \sum_j \frac{L_o^j}{L^j} M_{jd}^m \, \widetilde{\Delta R_d}, \qquad m \in \{\text{import}, \text{export}\}. \tag{5}$$

- M_{jd}^m is the baseline value of industry-j imports or exports between the Philippines and destination d.

- L_o^j/L^j apportions the national industry-destination trade shock to province o according to its baseline employment share in industry j.
- **Interpretation.** These variables proxy for province-level exposure to trade shocks induced by the same exchange-rate movements.
- **Finding.** Adding them to the regression leaves the migrant-income coefficient essentially unchanged.

3.7 Education channel in the model-based quantification

The paper's main text summarizes the model-based quantification, with full derivations delegated to Supplemental Appendix C. The logic is:

$$\Delta SkillShare_o = \widehat{\beta}^{edu} \cdot Shiftshare_o,$$

where $\widehat{\beta}^{edu}$ is the estimated college-completion response from Table 2.

- The change in the skilled share is fed through pre-shock skill-specific migration probabilities and pre-shock skill premia in both domestic and migrant earnings.
- **Interpretation.** A positive migrant-income shock relaxes liquidity constraints and raises returns to education, increasing skill acquisition; this in turn raises both migration opportunities and domestic earnings.
- **Quantitative result.** The education channel explains about 19.1% of the migrant-income increase, 19.9% of the domestic-income increase, and 19.7% of the global-income increase.

3.8 Migration gravity response in the structural framework

The model also allows migration flows to respond to destination wages, with elasticity θ:

$$\log M_{od} = \theta \log w_d + \text{other terms.}$$

- The paper estimates θ using PPML on origin-destination migration flows and destination wage shocks.
- **Estimated elasticity.** $\theta \approx 3.42$.
- **Interpretation.** Higher destination wages induced by persistent exchange-rate appreciations reallocate migration flows toward better-paying destinations, thereby magnifying long-run migrant income.
- **Quantitative result.** Exchange-rate persistence and migration reallocation explain an additional 69.9% of the migrant-income increase, on top of the education channel.

3.9 Plausibility of domestic-income magnitudes

The paper then asks whether the large domestic-income response can be rationalized by standard mechanisms. The discussion combines three ingredients:

- a share α of migrant income returned to origin provinces,
- a local demand multiplier,
- enterprise investment with high but declining returns.

- With $\alpha = 0.64$ and a multiplier around 2.9, the demand-multiplier channel alone explains most of the remaining domestic-income effect.
- Adding enterprise investment and declining returns, the model can generate about PhP 18.93 in domestic income by 2015 for a PhP 1 initial migrant-income shock, which is close to the estimated magnitude.
- **Interpretation.** The point is not to pin down a unique structural decomposition of domestic income, but to show that the reduced-form magnitudes are economically plausible under reasonable assumptions.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A simple regression of domestic income on migrant income would be contaminated by reverse causality and omitted variables.
- Richer provinces may both generate higher domestic income and send more migrants, and provinces with more educated populations may have both higher incomes and stronger migration opportunities.
- Exchange-rate shocks may also affect trade or prices, so one must isolate the migrant-income channel from alternative channels.

4.2 What identifies the model in the new approach

- **Exogenous shares logic.** Identification comes from pre-1997 province-destination migrant income shares, which determine exposure to destination-specific exchange-rate shocks during the Asian Financial Crisis.
- **Tailored shares.** The paper argues these shares are especially suited to the treatment of interest because they are specific to migrant income rather than generic industrial composition.
- **Lagged shares.** Shares are measured in 1995, two years before the crisis, which helps rule out anticipatory behavior.
- **Parallel-trend requirement.** Absent the exchange-rate shocks, provinces with different exposure weights would have evolved similarly after conditioning on baseline controls.

4.3 Why the identifying assumptions are plausible

- The 1997 Asian Financial Crisis was largely unanticipated.

- Pre-existing migration networks arose from historical labor demand abroad and migration costs linked to geography, language, and social networks, not from anticipation of the crisis.
- The paper directly tests for pre-trends up to 12 years before the crisis and finds no worrying evidence that high-exposure provinces were already on systematically different trajectories.

4.4 Why trade and FDI are unlikely confounders

- The migrant-income shift-share is not strongly correlated with the import and export shift-share controls after conditioning on the baseline covariates used in the main regressions.
- Manufactured exports do not respond significantly to the migrant-income shock.
- Agricultural income also does not respond, which argues against agricultural-export channels.
- Consumer prices do not rise in more exposed provinces, so the expenditure and income effects are not just local inflation.
- Country-level inward FDI to the Philippines is not significantly related to the destination exchange-rate shocks, making it unlikely that province-level FDI responses explain the results.

4.5 SUTVA, internal migration, and spatial spillovers

- A concern is that shocks to one province might spill over to neighboring provinces through trade or labor mobility.
- The paper finds no large or statistically significant effect on net internal migration, with only a small reduction in youth outmigration that cannot explain the main results.
- Controlling for the inverse-distance-weighted average shock in neighboring provinces does not materially affect the estimates.
- The authors argue that any residual spillovers are likely positive and therefore would bias estimates toward zero.

4.6 Persistence matters for interpretation

- The paper is explicitly about persistent rather than transitory migrant-income shocks.
- Both the exchange-rate component and the exposure-share component remain persistent well into the long run.
- This persistence is what allows the initial shock to generate a cumulative development process through education, migration reallocation, and domestic investment.

4.7 Relation to the literature

- The paper contributes to the migration-and-development literature by studying origin-area effects rather than only migrant or household effects.

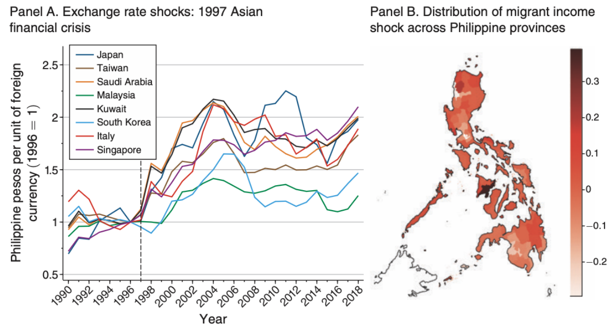

FIGURE 1

Notes: Left panel shows exchange rate changes for selected countries. Data are from World Development Indicators (World Bank 2024), annual average nominal exchange rates in units of foreign currency per Philippine peso, normalized to 1 in 1996, for 8 large sources of international migrant income for Philippine provinces. Vertical dashed line indicates 1997 (year of the Financial Crisis). The right panel shows the spatial variation in province-o shift-share variable (migrant income shock) $Shiftshare_o = MigInc_{o0}\ Rshock_o$ after partialling out weighted average exchange rate shock $Rshock_o$ and pre-shock migrant income per capita $MigInc_{o0}$, for 74 Philippine provinces. See Section IV and Supplemental Appendix Section C.2 for details.

- It also contributes to the modern econometric literature on shift-share designs by using an "exogenous shares" setup and by providing extensive balance and robustness checks.
- Relative to prior work on migration shocks, its distinct contribution is the long horizon, the simultaneous focus on migrant, domestic, and global income, and the combination of reduced-form evidence with structural quantification.

5 Tables and figures

5.1 Figure 1: Exchange-rate persistence and spatial variation in exposure

Read:

- Panel A shows that the destination-country exchange-rate shocks from the 1997 Asian Financial Crisis were highly persistent, not temporary blips.
- Panel B shows substantial cross-province variation in the shift-share treatment once one partials out the average exchange-rate shock and overall migrant-income exposure.
- This figure visually establishes the two ingredients the design needs: persistent shifts and heterogeneous baseline exposure across provinces.

5.2 Table 1 and Figure 2: Income and expenditure effects

Read:

- In panel D of Table 1, the coefficient on $Shiftshare_o \times Post_t$ is about 12.95 for domestic income and 12.33 for expenditure in the full triennial sample, and about 18.24 for domestic income, 6.12 for migrant income, 24.35 for global income, and 13.80 for expenditure in the long-run sample focused on 1994, 2009, 2012, and 2015.
- A one-standard-deviation shock raises domestic income per capita by about PhP 1,204 and expenditure by about PhP 1,147.

Table 1—Effects of Migrant Income Shock on Global Income, Domestic Income, Migrant Income, and Expenditure per Capita

	Triennial: 1985–2018		1994, 2009, 2012, and 2015			
	Domestic income per capita (1)	Expenditure per capita (2)	Global income per capita (3)	Domestic income per capita (4)	Migrant income per capita (5)	Expenditure per capita (6)
Panel A. Destination controls only						
$Shiftshare_o$ × Post	12.913	10.877	30.732	24.509	6.222	19.058
	(8.687)	(7.925)	(9.531)	(8.591)	(2.058)	(8.907)
Panel B. Additional province development status controls						
$Shiftshare_o$ × Post	9.562	9.779	24.241	18.173	6.068	13.302
	(6.148)	(5.055)	(6.264)	(6.128)	(1.683)	(5.188)
Panel C. Additional province industrial structure controls						
$Shiftshare_o$ × Post	13.154	12.442	24.499	18.348	6.151	13.859
	(4.887)	(4.638)	(6.149)	(6.033)	(1.612)	(5.386)
Panel D. Additional import and export shift-share variables						
$Shiftshare_o$ × Post	12.947	12.332	24.351	18.235	6.117	13.798
	(4.066)	(3.891)	(5.770)	(5.668)	(1.685)	(4.640)
Observations	813	813	296	296	296	296
Baseline DV mean	26.102	24.497	30.189	26.102	4.087	24.497
Baseline DV SD	9.406	8.734	11.400	9.406	2.993	8.734

Note: Unit of observation is the province-year. Domestic income and expenditure per capita are from Family Income and Expenditure Survey (FIES). Migrant income per capita is calculated from POEA/OWWA and Philippine census data. Global income per capita is migrant income per capita plus domestic income per capita. Income and expenditure are in thousands of real 2010 Philippine pesos (PhP17.8 per PPP US$ in 2010). The year 1997 is dropped from the analysis as the exchange rate shock takes place in 1997. Outcome data are not available for one province (Rizal) in 1988 due to a fire that destroyed survey records. Destination pre-shock controls are (all for 1995): GDP per capita of the destination; mean annual income per Philippine migrant in the destination; share of Philippine migrants to the destination working in professional occupations (highest-skilled general occupational category); share of Philippine migrants to the destination working in manufacturing occupations (intermediate-skilled general occupational category; the lowest-skilled general occupational category, services, is the omitted category); share of all Philippine migrants going to the destination; share of baseline migrant from province going to Middle East and North Africa; share of baseline migrant from province going to East Asia; share of baseline migrant from province going to Middle East and OECD countries. Destination country-level controls are aggregated to the province level using Borusyak, Hull, and Jaravel (2022) weights (province's pre-shock aggregate migrant income in the destination). Province development status pre-shock controls are as follows: share of households that are rural and household asset index (from 1990 census); domestic income per capita and expenditure per capita (average across 1988/1991/1994 FIES). Province industrial structure pre-shock controls are as follows: share of workforce in primary sector, share of workforce in manufacturing, share of workforce in service sector, share of workforce in financial and business services (from 1990 census). Baseline dependent variable mean and standard deviation calculated based on data from the pre-shock year nearest to the crisis. All regressions include province and year fixed effects. Standard errors are clustered at the province level.

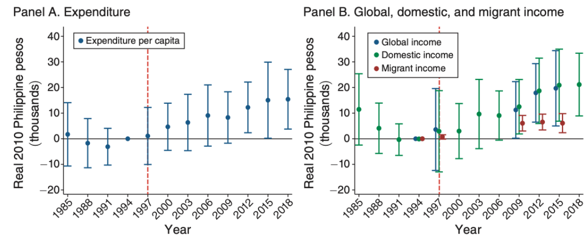

Figure 2. Event Studies for Expenditure and Income per Capita

Notes: Regressions modify equation (4) to include interactions between $Shiftshare_o$ and indicator variables for each pre- and post-shock year. The 1994 interaction term is omitted as reference point. Specification corresponds to that of Table 1, panel D (including province fixed effects, year fixed effects, and controls for differential trends with respect to pre-shock province characteristics, destination characteristics, and province import and export shift-share variables). Expenditure per capita includes food, education, durable goods, and housing, among other categories. Domestic income per capita includes earned income from wage and entrepreneurial activities, along with income from all other sources excluding transfers from abroad and domestic sources. Migrant income per capita is the sum of all income earned outside the Philippines by a province's migrants. Global income per capita is the sum of domestic and migrant income per capita. Outcomes are in real 2010 Philippine pesos (PhP17.8/US$ PPP). Observations are at the province-period level, and include each triennial period between 1985 and 2018 inclusive (when available); unlike in Table 1, we now include partially treated year 1997 in the sample. 95 percent confidence intervals shown. Standard errors are clustered at the province level.

- The event studies in Figure 2 show no troubling positive pre-trends and instead gradual post-1997 accumulation of effects.
- The large implication is that the shock does not merely raise overseas income; it also produces sizable and persistent local development gains.

5.3 Table 2: Education

Read:

- The effects on primary schooling are essentially zero, but the effects on secondary and college completion are positive and robust.
- In panel D, the coefficients are about 0.059 for secondary completion and 0.045 for college completion.
- The paper translates these into about 0.55 percentage points more secondary completion and 0.42 percentage points more college completion for a one-standard-deviation shock.
- Education is therefore a central long-run mechanism rather than a side result.

5.4 Table 3: Migration rates, migrant skill, and occupational upgrading

Read:

- The migrant-income shock raises the migrant share of the population, the share of migrants who are skilled, and average migrant salary.

TABLE 2—EFFECTS OF MIGRANT INCOME SHOCK ON EDUCATION

	Share completed		
	Primary school (1)	Secondary school (2)	College (3)
Panel A. Destination controls only			
Shiftshare$_o$ × Post	−0.006	0.081	0.037
	(0.029)	(0.033)	(0.016)
Panel B. Additional province development status controls			
Shiftshare$_o$ × Post	−0.007	0.051	0.051
	(0.036)	(0.029)	(0.018)
Panel C. Additional province industrial structure controls			
Shiftshare$_o$ × Post	0.004	0.058	0.045
	(0.034)	(0.028)	(0.016)
Panel D. Additional import and export shift-share variables			
Shiftshare$_o$ × Post	0.005	0.059	0.045
	(0.025)	(0.028)	(0.015)
Observations	444	444	444
Baseline DV mean	0.763	0.419	0.122
Baseline DV standard deviation	0.104	0.114	0.039

Notes: Unit of observation is the province-year. Analysis uses census data; periods are 1990, 1995, 2000, 2007, 2010, and 2015. Dependent variables are share of population (aged 20–64) who have completed primary, secondary (high school), and college education. Primary school, secondary school, and college completion is defined as having completed at least 6, 10, and 14 years of schooling respectively. For list of destination and provincial controls, see Table 1. Baseline dependent variable mean and standard deviation calculated based on data from nearest pre-shock year. All regressions include province and year fixed effects. Standard errors are clustered at the province level.

TABLE 3—EFFECTS ON MIGRATION RATE, MIGRANT SKILL, AND CONTRACT TYPE

	Census			Contracts per 10,000 working age people			
	Migrant share age 20–64 (1)	Share skilled migrants (2)	Average mig. salary (3)	First qtile education (4)	Second qtile education (5)	Third qtile education (6)	Fourth qtile education (7)
Panel A. Destination controls only							
Shiftshare$_o$ × Post	0.018	0.176	171.847	57.729	3.810	75.001	52.223
	(0.006)	(0.046)	(159.574)	(74.162)	(6.908)	(30.078)	(35.411)
Panel B. Additional province development status controls							
Shiftshare$_o$ × Post	0.018	0.214	241.279	38.617	−0.229	56.869	19.017
	(0.007)	(0.061)	(155.849)	(63.775)	(6.243)	(22.641)	(24.129)
Panel C. Additional province industrial structure controls							
Shiftshare$_o$ × Post	0.018	0.188	254.343	36.237	−0.806	50.785	16.298
	(0.006)	(0.054)	(153.234)	(61.614)	(5.815)	(17.862)	(21.005)
Panel D. Additional import and export shift-share variables							
Shiftshare$_o$ × Post	0.018	0.187	251.146	35.464	−0.883	50.359	15.933
	(0.006)	(0.055)	(138.529)	(60.839)	(6.260)	(18.142)	(21.611)
Observations	444	444	296	296	296	296	296
Baseline DV mean	0.011	0.300	431.718	52.103	6.468	18.324	18.664
Baseline DV SD	0.008	0.089	145.806	45.834	7.934	28.371	24.710

Notes: Unit of observation is the province-year. Migrant share in the population and share of migrant workers who are skilled is from the census (periods are 1990, 1995, 2000, 2007, 2010, and 2015). Skilled is defined as completing 14 years of education, which corresponds to finishing a college degree. Migrant contract variables are calculated from POEA/OWWA data (periods are 1994, 2009, 2012, and 2015). Outcome variable in column 3 is average annual migrant salary and columns 4–7 are the migrant contracts (per 10,000 working age population) in occupations in the first (lowest) through fourth (highest) quartiles of migrant years of education. Average annual migrant salary is in thousands of real 2010 Philippine pesos (PhP17.8 per PPP US$ in 2010). For list of destination and provincial controls, see Table 1. Baseline dependent variable mean and standard deviation calculated based on data from nearest pre-shock year. All regressions include province and year fixed effects. Standard errors are clustered at the province level.

- In panel D, the coefficients are about 0.018 for migrant share, 0.187 for skilled migrant share, and 251 thousand real pesos for average migrant salary.
- New migrant contracts rise mainly in the higher-education occupation quartiles, especially the third quartile, rather than in the lowest-skill occupations.
- The paper's interpretation is that provinces exposed to better migrant-income opportunities do not just send more migrants; they send more skilled migrants into better-paying jobs.

5.5 Table 4: Components of domestic income

Read:

- The domestic-income gains come primarily through wage income and entrepreneurial/rental income, not through passive "other" income.
- In panel D, wage income rises by about 10.46, entrepreneurial and rental income by about 7.27, and other income is essentially unchanged.
- This pattern supports the view that the shock stimulates actual productive activity at origin rather than merely generating accounting transfers.

5.6 Table 5: Structural change

Read:

- Domestic income gains are concentrated in nonagricultural activities: panel D shows little effect on agricultural income but a large positive effect (about 15.54) on nonagricultural income.
- Within entrepreneurial income, the service sector contributes the most; service entrepreneurial income rises much more than primary or manufacturing entrepreneurial income.

TABLE 4—EFFECTS OF MIGRANT INCOME SHOCK ON COMPONENTS OF DOMESTIC INCOME

	Domestic income components		
	Wage income (1)	Entrepreneurial and rental income (2)	Other income (3)
Panel A. Destination controls only			
Shiftshare$_o$ × Post	12.949	9.770	1.790
	(5.256)	(4.248)	(1.806)
Panel B. Additional province development status controls			
Shiftshare$_o$ × Post	10.959	7.398	−0.184
	(4.515)	(3.753)	(2.263)
Panel C. Additional province industrial structure controls			
Shiftshare$_o$ × Post	10.545	7.273	0.530
	(4.082)	(3.834)	(2.069)
Panel D. Additional import and export shift-share variables			
Shiftshare$_o$ × Post	10.461	7.266	0.508
	(4.159)	(3.545)	(2.082)
Observations	296	296	296
Baseline DV mean	11.759	9.987	4.355
Baseline DV standard deviation	6.789	3.109	2.002

Notes: Unit of observation is the province-year. Data from the Family Income and Expenditure Survey (FIES); periods are 1994, 2009, 2012, and 2015. For list of destination and provincial controls, see Table 1. Baseline dependent variable mean and standard deviation calculated based on data from nearest pre-shock year. All regressions include province and year fixed effects. Standard errors are clustered at the province level.

TABLE 5—EFFECTS OF MIGRANT INCOME SHOCK ON STRUCTURAL CHANGE

	Overall income (FIES)		Entrepreneurial income (FIES)			Employment share (census)		
	Agricultural (1)	Non agricultural (2)	Primary (3)	Manufacturing (4)	Services (5)	Primary (6)	Manufacturing (7)	Services (8)
Panel A. Destination controls only								
Shiftshare$_o$ × Post	1.714	22.795	1.637	1.833	5.705	−0.139	0.061	0.078
	(2.847)	(8.402)	(2.835)	(0.918)	(2.557)	(0.045)	(0.028)	(0.048)
Panel B. Additional province development status controls								
Shiftshare$_o$ × Post	2.534	15.639	1.528	1.490	3.866	−0.133	0.065	0.068
	(3.000)	(5.690)	(3.178)	(0.918)	(1.889)	(0.044)	(0.029)	(0.053)
Panel C. Additional province industrial structure controls								
Shiftshare$_o$ × Post	2.670	15.678	1.766	1.366	3.630	−0.125	0.034	0.091
	(2.760)	(5.398)	(3.107)	(0.874)	(1.764)	(0.045)	(0.020)	(0.038)
Panel D. Additional import and export shift-share variables								
Shiftshare$_o$ × Post	2.694	15.540	1.787	1.358	3.609	−0.126	0.034	0.092
	(2.629)	(5.362)	(2.970)	(0.865)	(2.008)	(0.046)	(0.021)	(0.038)
Observations	296	296	296	296	296	296	296	296
Baseline DV mean	7.759	18.343	5.712	0.550	3.561	0.559	0.051	0.390
Baseline DV SD	3.074	10.534	2.826	0.556	2.059	0.185	0.051	0.141

Notes: Unit of observation is the province-year. Data from the Family Income and Expenditure Survey (FIES) and census. FIES analysis includes years 1994, 2009, 2012, and 2015. Census analysis includes years 1990, 1995, 2000, 2010. Services are broadly defined to include the nontradable utilities and construction sectors. For list of destination and provincial controls, see Table 1. Baseline dependent variable mean and standard deviation calculated based on data from nearest pre-shock year. All regressions include province and year fixed effects. Standard errors are clustered at the province level. All regressions include province and year fixed effects.

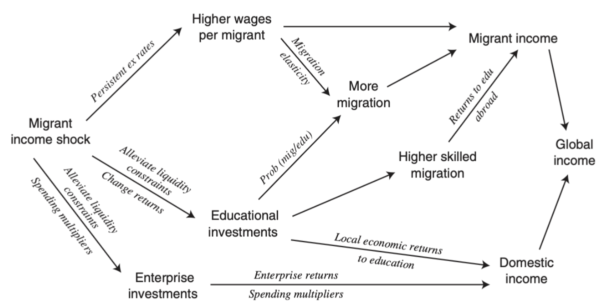

FIGURE 3. STYLIZED OVERVIEW OF POSSIBLE CHANNELS

Notes: Overview of modelled channels via which the migrant income shock affects global income. Details in Supplemental Appendix C.

- Employment shifts out of primary sectors and toward nontradable goods and services. In panel D, the primary employment share falls by about 0.126, while the services share rises by about 0.092.
- This is strong evidence that migrant-income shocks induce broader structural transformation, not just more income in the same local production structure.

5.7 Figure 3 and model-based quantification

Read:

- Figure 3 lays out the paper's channel decomposition: persistent exchange-rate gains, education, migration reallocation, spending multipliers, and enterprise investment jointly map the initial migrant-income shock into long-run global income gains.
- The education channel explains about one-fifth of total long-run global-income gains.
- Education plus persistent exchange-rate gains and migration reallocation explain about 89% of the migrant-income increase.
- A simple multiplier-and-investment accounting exercise shows that the large domestic-income

response is quantitatively plausible under reasonable assumptions.

6 Short summary

- **Conclusion.** Persistent improvements in overseas migrant income prospects generate large and lasting development gains in migrant-origin provinces.
- **Identification insight.** The key source of exogenous variation is a province-level migrant-income shift-share shock built from pre-crisis destination exposure and destination-specific exchange-rate movements during the 1997 Asian Financial Crisis.
- **Main economic lesson.** Most of the long-run gains come from domestic income, not just migrant income. The shock raises education, migration quality, and local structural transformation, creating a long-run development process.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that migration opportunities can be a development shock, not merely a household remittance shock.
- Persistent access to high-income overseas labor markets raises local education, skilled migration, domestic wage and entrepreneurial income, and shifts labor away from primary sectors.
- The quantitative results imply that migration policy belongs in the core development-policy toolkit.

7.2 Economic intuition

- A temporary windfall would mainly create a level effect. This paper instead studies a *persistent* improvement in migrant earning opportunities.
- Because the shock persists, households have reason to invest in skill acquisition, and migration networks reallocate workers toward better-paying destinations.
- Those higher migrant earnings then feed back into the origin economy through spending, enterprise investment, and structural change, so domestic income becomes the dominant component of total long-run gains.

7.3 Takeaway

This paper's central message is that the developmental payoff from migration is much broader than the direct income gain accruing to migrants themselves. When migrant income opportunities improve persistently, origin areas can become richer mostly through domestic channels: more schooling, more productive migration, more entrepreneurial activity, and a structural shift toward nonagricultural local production. In that sense, the paper reframes migration policy as a long-run development policy.